

Statistical Analysis Plan

Does Cognitive Outsourcing to Generative AI Impair Skill Acquisition?

A Three-Arm Randomised Controlled Trial of LLM Scaffolding in
Visualisation Literacy

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This document is an extract from the pre-registration analysis plan of the parent thesis manuscript. It reproduces, verbatim, the Statistical Analysis Plan (main text) and its Extended Justification appendix, built directly from the live section sources (`sections/SAP_04_07.tex`, `sections/SAP_appendix_04_07.tex`), which are `\input` from the parent manuscript's Methods and Appendix chapters. This version supersedes the earlier extract built from the `drafts/` review copies and incorporates the wording revisions made directly to the live source since that build. Cross-references to the Power Analysis appendix (`app:power`), a table in the Theory chapter (`tab:clt-patterns`), and a Theory-chapter section (`sec:theory`) point to material not reproduced in this extract; they render as “??” below and are not a defect introduced by this extract.

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0.1 Statistical Analysis Plan

Identification strategy. The primary estimand is specified with intent-to-treat (International Conference on Harmonisation, 1998), analysing the effect of being *assigned* to a tool condition, irrespective of how intensively or faithfully the tool was used. All randomised participants who provide outcome data are analysed in the arm to which they were randomised, as required for the confirmatory analysis of a parallel-group trial under CONSORT (Hopewell et al., 2025). Intercurrent events, namely use of external tools during training (relevant for the search arm) and post test (disclosed via the end-of-study item) and minimal engagement with the assigned tool, are handled under the treatment-policy strategy in the primary analysis; complier-specific estimands are reported as supplementary analyses (see Compliance below). One qualification applies as H_{2a} is a non-inferiority claim, for which intention-to-treat is not conservative, we therefore also discuss the per-protocol analysis as recommended by the CONSORT extension for non-inferiority trials (Piaggio et al., 2012) (see Hypothesis testing and Compliance).

Measurement and outcome hierarchy. Two outcomes are measured on every block (training and both unaided post-tests), with a pre-specified hierarchy between them:

1. **Accuracy** (*primary outcome*): the proportion of correct responses on each 16-item unassisted block;
2. **Success time** (*key secondary outcome*): the per-item time elapsed from item onset to the submission of a *correct* answer. Items answered incorrectly, answered with “I do not know”, or left unanswered are treated as right-censored at their recorded response time, so that every administered item contributes information.

The censoring formulation is essential: restricting a response-time analysis to correctly answered items would condition on a post-treatment variable, since correctness is itself affected by the manipulation, and such conditioning can distort treatment-effect estimates (Montgomery et al., 2018). The pair is theoretically motivated: accuracy indexes schema *construction* and success time schema *automation*, the two CLT phases of skill acquisition, so both dimensions of the construct are measured and reported. The confirmatory decision, however, is carried by accuracy alone: a single pre-specified primary variable keeps

the confirmatory claim unambiguous and concentrates statistical power where the theoretical prediction is sharpest (International Conference on Harmonisation, 1998; Pocock et al., 1987). The full defence of the outcome pair and of this hierarchy, including the rejected co-primary and composite alternatives, is given in Appendix 0.2.1.

Primary analysis: accuracy. Because each participant is measured at both unassisted waves, the two proportions correct are correlated within person; accuracy is therefore analysed with a participant-level linear mixed model (REML) (Detry & Ma, 2016), whose random intercept absorbs that dependence and which yields, from a single fit, the immediate-wave arm effects (the primary estimand) together with their change at the delayed wave (the persistence estimand):

$$Y_{iw} = \beta_0 + \beta_1 \text{Socratic}_i + \beta_2 \text{Unrestricted}_i + \delta \text{Wave}_w + \beta_4 (\text{Socratic}_i \times \text{Wave}_w) + \beta_5 (\text{Unrestricted}_i \times \text{Wave}_w) + u_i \quad (1)$$

with Google-only the reference category and Wave a 0/1 dummy (immediate = 0, delayed = 1), so that every arm coefficient is anchored at the immediate post-test. The parameters map onto the hypotheses as follows. β_2 , the immediate-wave Unrestricted–Google difference, adjudicates H1_a ($H_0: \beta_2 \geq 0$ against $H_1: \beta_2 < 0$). β_1 , the immediate-wave Socratic–Google difference, adjudicates the non-inferiority hypothesis H2_a ($H_0: \beta_1 \leq -\Delta$ against $H_1: \beta_1 > -\Delta$, for the pre-specified margin Δ ; see Hypothesis testing). The linear combination $\beta_1 - \beta_2$ adjudicates the supporting Socratic–Unrestricted contrast ($H_0: \beta_1 - \beta_2 \leq 0$ against $H_1: \beta_1 - \beta_2 > 0$). Persistence (H5_a) is the claim that the H1_a deficit is still present at 72 hours; it is therefore adjudicated by the *delayed-wave simple effect* $\beta_2 + \beta_5$ ($H_0: \beta_2 + \beta_5 \geq 0$ against $H_1: \beta_2 + \beta_5 < 0$), while the interaction β_5 itself estimates how much the deficit grows or shrinks between waves and is reported with its confidence interval rather than tested as the persistence claim. Here u_i is a participant random intercept and superscript c denotes the mean-centred prognostic baseline covariates (MiniVLAT, AIUse), adjusted for precision; single-wave linear models elsewhere in this plan use HC3 robust standard errors. A practical property of this anchoring is that the primary contrasts are measured in the same session as training, before the attrition window opens, so they are essentially complete-data by design. The choice of the mixed

model over per-wave analyses and repeated-measures ANOVA, the covariate adjustment, the compound-symmetry restriction and its guards, and the assumption diagnostics are detailed in Appendix 0.2.2.

Key secondary analysis: success time. Not every item yields a correct answer, so the time to a correct response is right-censored whenever the answer is wrong or “I do not know”. It is therefore analysed as a censored failure time with a Cox proportional hazards model (Cox, 1972), stratified by wave so that each wave keeps its own baseline hazard, with wave-specific arm effects:

$$\lambda_w(t \mid \cdot) = \lambda_{0w}(t) \exp(\theta_1 \text{Socratic}_i + \theta_2 \text{Unrestricted}_i + \theta_3 (\text{Socratic}_i \times \text{Wave}_w) + \theta_4 (\text{Unrestricted}_i \times \text{Wave}_w)) \quad (2)$$

where the event is the submission of a correct answer; incorrect, “I do not know”, and unanswered items are censored at their recorded response time; and Wave_w is the same 0/1 dummy as in Equation (1), its main effect absorbed by the strata. A hazard ratio above one indicates a higher instantaneous rate of producing correct answers—that is, greater fluency. With this coding θ_2 is the immediate-wave log hazard ratio of Unrestricted against Google and adjudicates H1_b ($H_0: \theta_2 \geq 0$, i.e. $\text{HR} \geq 1$, against $H_1: \theta_2 < 0$); the combination $\theta_1 - \theta_2$ adjudicates the supporting Socratic–Unrestricted fluency contrast of H2_b ($H_0: \theta_1 - \theta_2 \leq 0$); and the delayed-wave combinations $\theta_1 + \theta_3$ and $\theta_2 + \theta_4$ provide the exploratory success-time persistence read (H5_b), reported as estimates with confidence intervals only. No non-inferiority claim is made on the time scale. Because items are nested in participants, inference uses participant-clustered robust standard errors. Success time is a key secondary outcome: analysed and reported with the same rigour as accuracy, but it does not gate the confirmatory decision. Estimator validity under clustering, the handling of ties, the proportional-hazards check and its accelerated-failure-time fallback, the censoring cap, and the competing-risks rationale for this estimand are detailed in Appendix 0.2.3.

Hypothesis testing and multiplicity. All confirmatory hypotheses are directional and tested one-sided at $\alpha = 0.05$; two-sided 95% confidence intervals accompany every

estimate. The confirmatory family concerns the primary outcome only and is tested with a fixed-sequence gatekeeping procedure (Bauer, 1991; Dmitrienko & D’Agostino, 2013; Maurer et al., 1995), which controls the familywise error rate in the strong sense while testing each hypothesis at the full α :

1. **H1_a (superiority)**. Reject $H_0: \beta_2 \geq 0$ if the one-sided $p < 0.05$.
2. **H2_a (non-inferiority)**. Tested only if step 1 rejects. Non-inferiority is declared if the lower bound of the two-sided 90% confidence interval for β_1 (equivalently, the one-sided 95% bound) exceeds $-\Delta$ (Lakens, 2017; Lakens et al., 2018; Schuirmann, 1987). The margin is pre-specified at $\Delta = 8.5$ percentage points, defined by 50% preservation of the ≈ 17 -percentage-point reference deficit that anchors H1 (Shen & Tamkin, 2026): the Socratic arm must retain at least half of the control’s advantage over the harmful comparator. The margin is justified on substantive rather than power grounds (Althunian et al., 2017; D’Agostino et al., 2003; European Medicines Agency, 2005; Piaggio et al., 2012); its derivation and the non-inferiority power analysis are given in Appendix ???. That same reporting guidance additionally requires the claim to hold in the per-protocol population (Piaggio et al., 2012), and the 90% interval is reported whatever the verdict (Cumming, 2014).
3. **Supporting contrast**. Tested only if step 2 passes: reject $H_0: \beta_1 - \beta_2 \leq 0$ if the one-sided $p < 0.05$ (Socratic superior to Unrestricted—the contrast establishing that the scaffold, not mere LLM presence, protects learning).

Testing stops at the first non-rejection; any hypotheses beyond that point are reported as estimates with confidence intervals, without confirmatory claims. Two further pre-specified secondary hypotheses on the accuracy scale—the Bloom-order interaction (H4_a, Equation (3)) and delayed-wave persistence (H5_a, Equation (1))—form a separate family tested one-sided with the Holm step-down procedure at familywise $\alpha = 0.05$ (Holm, 1979). The two directional key-secondary success-time contrasts (H1_b and the H2_b supporting contrast) form a third, Holm-adjusted family interpreted as supporting evidence only; the Socratic–Google hazard ratio, for which no directional hypothesis is specified, accompanies them as an estimate with its confidence interval. The aided-block productivity indicators

(H3) are controlled at a false discovery rate as converging, non-confirmatory indicators (Benjamini & Hochberg, 1995). All remaining quantities are exploratory and carry no error-rate claim; these include the arm-by-Mini-VLAT moderation of the immediate-wave accuracy deficit motivated in Section ??, fitted by adding arm-by-covariate interaction terms to Equation (1) and reported as an estimate with its confidence interval, without a directional claim or verdict. The choice of the gatekeeper over a flat Holm correction and over graphical α -recycling, and the full non-inferiority machinery, are defended in Appendix 0.2.4.

Secondary analysis I: Bloom-level concentration. To test RQ2, the focal analysis contrasts lower-order (Remember, Understand) against higher-order (Analyze, Evaluate) accuracy on the immediate unaided post-test, modelling item-level correctness with a logistic mixed model with crossed participant and item random intercepts (Baayen et al., 2008; Barr et al., 2013):

$$\text{logit}(\Pr[Y_{im} = 1]) = \beta_0 + \beta_1 \text{Socratic}_i + \beta_2 \text{Unrestricted}_i + \beta_3 \text{Order}_m + \beta_4 (\text{Socratic}_i \times \text{Order}_m) + \beta_5 (\text{Unres}$$
(3)

where Y_{im} is the correctness of participant i on item m , $\text{Order}_m = 1$ for higher-order items and 0 for lower-order, and u_i and v_m are crossed participant and item random intercepts. The focal, directional quantity is the arm $\times$ order interaction β_5 for the Unrestricted–Search contrast—the item-level expression of H4_a ($H_0: \beta_5 \geq 0$ against $H_1: \beta_5 < 0$ on the log-odds scale): a negative value denotes a deficit larger at the higher-order level. It is tested within the secondary Holm family (see Hypothesis testing and multiplicity), with the confirmatory one-sided test taken from the participant-clustered logistic GEE fit of this mean structure (see Software and reproducibility; Appendix 0.2.5). The success-time analogue H4_b is exploratory, reported as order-specific estimates with confidence intervals only. The preference for this binary partition over a four-level analysis, the supportive four-level trend, the exploratory wave extension and lower-order equivalence test, and the validity check on the level ordering are detailed in Appendix 0.2.5.

Secondary analysis II: productivity. The aided practice block measures the productivity side of the trade-off and adjudicates, in supporting fashion, the assisted-performance predictions $H3_a$ (accuracy) and $H3_b$ (success time). Each participant contributes one aided-block value per outcome, so aided accuracy is analysed in the covariate-adjusted single-wave framework

$$Y_i = \beta_0 + \beta_1 \text{Socratic}_i + \beta_2 \text{Unrestricted}_i + \gamma_1 \text{MiniVLAT}_i^c + \gamma_2 \text{AIUse}_i^c + \varepsilon_i, \quad (4)$$

fitted by ordinary least squares with HC3 standard errors; aided success time uses the Cox model of Equation (2) restricted to the training block, without wave strata or wave interactions. In both, β_2 (respectively the aided-block θ_2) evaluates $H3$ ($H_0: \beta_2 \leq 0$ against $H_1: \beta_2 > 0$: tool access raises assisted performance). These outcomes are converging indicators rather than confirmatory claims and are controlled at a false discovery rate (Benjamini & Hochberg, 1995); they double as the affordance manipulation check. The descriptive productivity–learning frontier plot and the rationale for the rate correct score are detailed in Appendix 0.2.6.

Robustness analyses. Each test is subjected to multiple robustness tests detailed in Appendix 0.2.7.

Compliance, contamination, and complier effects. The primary analysis retains all randomised participants regardless of compliance, because exclusions based on post-randomisation behaviour can bias the causal estimate (Montgomery et al., 2018). Two supplementary analyses characterise the effect of actual tool use: a descriptive per-protocol estimate excluding participants who disclosed external LLM or AI-overview use during training, and a complier average causal effect estimated by two-stage least squares with random assignment as the instrument for compliant tool use, where the instrumented exposure is engagement with the *assigned* tool, so that for each LLM-versus-Google contrast the first stage is the treated arm’s engagement rate (a one-sided-noncompliance design, the control arm having no access to the assigned tool) (Angrist et al., 1996), identified under the standard exclusion and monotonicity assumptions, with compliance

rates reported per arm. The per-protocol estimate additionally serves as the pre-specified co-analysis for the non-inferiority hypothesis H_{2a} (see Hypothesis testing and multiplicity): non-inferiority is claimed only if both the intent-to-treat and per-protocol estimates support it (Piaggio et al., 2012).

Missing data and attrition. Attrition is expected primarily at the three-day follow-up. Because the primary contrasts are anchored at the immediate wave, collected in the same session as training, attrition bears chiefly on the persistence estimand (H_{5a}). Differential attrition is assessed by comparing wave-2 return rates across arms and testing whether baseline covariates predict dropout differentially by arm. The mixed model of Equation (1) already uses all available data and is valid under missing-at-random (Detry & Ma, 2016); as sensitivity, the accuracy and success-time models are re-estimated on multiply imputed data using chained equations with $m = 50$ imputations including arm, baseline covariates, and wave-1 outcomes, pooled by Rubin’s rules (White et al., 2011). A pre-registration Monte Carlo validation of this multiple-imputation configuration under missing-at-random recovered an attenuated estimate of the delayed-wave contrast (-0.060 against a simulated truth of -0.080), whereas the primary mixed model of Equation (1) was essentially unbiased under the same design ($+0.001$ from truth). This plan therefore pre-commits to interpreting any discrepancy between the mixed-model and multiple-imputation estimates in favour of the mixed model, citing this diagnostic as the reason; the multiple-imputation estimate is reported as the pre-specified sensitivity analysis but never supersedes the primary conclusion. Worst-case sensitivity to attrition that violates missing-at-random is quantified with Lee (2009) bounds on the delayed-wave contrast, trimming the arm with lower attrition to bound the effect among always-responders without exclusion restrictions. Item-level missingness within completed blocks follows the pre-registered scoring key (unanswered items scored incorrect for accuracy; censored for success time).

Baseline reporting. Baseline characteristics are reported descriptively by arm with standardised mean differences; consistent with Senn (1994) and CONSORT (Hopewell et al., 2025), no hypothesis tests of baseline differences are performed.

Manipulation and implementation checks. Multiple pre-specified checks verify that the arms instantiated the intended manipulation; following Montgomery et al. (2018), their results inform interpretation and exploratory moderation but never trigger outcome-based exclusion of participants.

1. **Cognitive-load gradient.** Self-reported load is measured with the differentiated cognitive-load questionnaire of Leppink et al. (2013), whose ten items resolve into separate intrinsic-, extraneous-, and germane-load subscales rather than the single global index returned by an undifferentiated workload measure such as the NASA-TLX (Hart & Staveland, 1988). The differentiated instrument is required here because the offloading account predicts *component-specific* rather than global differences (Table ??): the germane subscale, which indexes the schema-construction effort that constitutes learning, is predicted to be depressed in the Unrestricted (outsourcing) arm relative to both the Socratic and the Control arm, whereas the extraneous subscale is predicted to be highest in the search-based Control arm. The instrument’s three-factor structure was established by confirmatory factor analysis and replicated across content domains and instructional contexts in its development studies (Leppink et al., 2013; Leppink et al., 2014), and it has since been independently validated, with acceptable subscale reliabilities, in problem-based and technology-mediated learning settings (Andersen & Makransky, 2021; Hadie & Yusoff, 2016). This differentiated approach is established in closely comparable research on AI-assisted learning: Zhang and Jiang (2026) use the Leppink instrument to link generative-AI dependency to the three load types, and Stadler et al. (2024) contrast an LLM with web search using the sibling scale of Klepsch et al. (2017). This subscale separation is precisely what renders the per-component predictions of Table ?? testable, which a single workload score cannot. Each subscale is compared across arms in the covariate-adjusted framework, and, for the components for which the theory specifies a monotone ordering of the arms, the contrast is supplemented by a Jonckheere–Terpstra test (Jonckheere, 1954) against the pre-registered ordered alternative. A germane-load gradient in which the Unrestricted arm falls below both comparison arms is the central self-report evidence that the arms elicited different

degrees of cognitive offloading.

2. **Engagement.** Logged tool interactions (search queries; chat messages sent) are summarised per arm
3. **Answer-adoption latency.** For every aided item the latency Δ_{ij} is the interval between the last tool interaction and the moment the participant logs their answer: for the two LLM arms the timestamp of the final assistant message, and—defining the parallel Google-arm analogue—for the search arm the timestamp of the last results-page interaction (query issued or result clicked). Both endpoints are read from the single shared wall-clock event stream of the interaction logger, so Δ_{ij} is well defined for all three arms without additional instrumentation. Short latencies indicate adoption of the tool’s output without independent verification, the behavioural signature of full outsourcing; longer latencies indicate deliberation. Because the appropriate cutoff for “direct adoption” is not known a priori, the check’s primary read is threshold-free: the per-arm Δ distribution is reported, and $\log\Delta$ is modelled with a linear mixed model with participant random intercepts contrasting Unrestricted against Socratic, the Unrestricted arm predicted to show shorter latencies, consistent with the offloading gradient (Table ??). A participant-level *direct-adoption rate*—the share of aided items with Δ_{ij} below a cutoff *locked from the pilot Δ distribution*—is then reported per arm as a descriptive secondary and carried into the exploratory moderation analysis (adoption rate $\times$ unassisted performance). Items with no tool interaction are excluded and their count reported.
4. **Socratic fidelity.** For the Socratic arm, the rate of judge-triggered regenerations is reported, and a random [20]% sample of assistant turns is independently double-coded by two human raters for answer disclosure, with inter-rater reliability quantified by Cohen’s κ (cohen1960) and a pre-registered adequacy threshold of $\kappa \geq 0.80$; the resulting disclosure rate quantifies residual contamination of the Socratic manipulation.

Internal consistency (KR-20) of each 16-item block is reported descriptively.

Software and reproducibility. All analyses are conducted in Python 3.11 with a pinned environment and fixed random seed. Estimation uses `pandas/numpy` for data handling; `statsmodels` for linear models with HC3 covariance (`cov_type="HC3"`), the linear mixed models of Equations (1) (REML via `MixedLM`, with a pre-specified optimiser fallback chain `lbfgs`→`bfgs`→`powell`→`cg` invoked when a near-zero random-intercept variance would otherwise leave the default solver singular), the fractional logit as a binomial GLM with robust quasi-maximum-likelihood covariance, multiple imputation via its MICE implementation, and, for the Bloom interaction of Equation (3), a participant-clustered logistic GEE (`cov_struct=Exchangeable`) as the confirmatory estimator, with the crossed-random-effects `BinomialBayesMixedGLM` retained only as corroboration because its variational posterior standard deviations are anticonservative; `lifelines` for the Cox models with participant-clustered robust covariance (the D. Y. Lin and Wei (1989) sandwich; because `lifelines`' `cluster_col` exposes robust standard errors per coefficient but keeps the naive covariance *matrix*, every `arm`×`wave` contrast is formed from the full cluster-robust covariance matrix reconstructed from the clustered score residuals), the proportional-hazards diagnostic of Grambsch and Therneau (1994) (`proportional_hazard_test`), and the log-normal accelerated failure time fallback; `scipy/numpy` for the permutation-based randomisation inference and a permutation implementation of the Jonckheere–Terpstra test; and `pingouin` for Hedges' g . Lee bounds are implemented directly from the trimming formulas of Lee (2009). The answer-adoption-latency $\log\Delta$ contrast uses the same participant-random-intercept `MixedLM`, with a participant-cluster-robust OLS fallback invoked when the random-effect variance is singular; the Δ values themselves are computed from the shared wall-clock ISO event timestamps carried through the interaction consolidation transform (`pcp_consolidate.js`), never from the model-generation latency. Multiplicity adjustments (Holm; Benjamini–Hochberg) use `statsmodels.stats.multitest`, and the non-inferiority bound for $H2_a$ is computed directly from the mixed-model contrast. Analysis code is written against the pre-registration before unblinding of outcome data, and code and de-identified data are archived in accordance with the pre-registration.

0.2 Statistical Analysis: Extended Justification

0.2.1 The outcome pair and its hierarchy: primary accuracy, key-secondary success time

The trial records two outcomes because the theory defines the construct as two-dimensional. CLT decomposes skill acquisition into schema *construction*—whether a correct response can be produced at all—and schema *automation*—the fluency with which it is produced (Kalyuga et al., 2003; Sweller et al., 1998). Accuracy indexes construction; success time indexes automation. The classical skill-acquisition literature makes the same decomposition, defining acquisition as joint improvement in accuracy *and* speed (Fitts & Posner, 1967; Proctor & Dutta, 1995). Measuring both therefore operationalises the whole construct rather than truncating it to either dimension. The two measures are also mutually constraining: because speed and accuracy trade off against one another (Heitz, 2014), neither is interpretable in isolation—an arm may appear accurate only by responding slowly and cautiously, or fast only by guessing. Anchoring time on *correct* responses collapses the two dimensions into an index of correct-response efficiency that fast guessing cannot inflate (Vandierendonck, 2017), neutralising this strategic confound. The pair is finally complementary along the Bloom gradient: lower-order items risk accuracy ceilings, where the compressed accuracy distribution loses discriminating power while success time remains continuous; on higher-order items, where few succeed and times are heavily censored, accuracy carries the signal.

Measuring both, however, does not require confirming on both. The confirmatory decision is carried by accuracy alone, with success time designated a key secondary outcome, for three reasons. First, a single pre-specified primary variable is the standard for a confirmatory trial (International Conference on Harmonisation, 1998); multiple primary endpoints require an explicit rule for what counts as trial success and an inferential architecture to match (Pocock et al., 1987). Second, construction is theoretically prior to automation: a schema that was never constructed cannot be automated, so the sharpest and least model-dependent prediction concerns accuracy, and the automation prediction presupposes the construction one. Third, the alternatives are strictly worse at this sample size. Co-primary endpoints in the regulatory sense (both must succeed) control the type I

rate without adjustment but compound the type II rate—the trial fails unless two effects are detected simultaneously—a material cost at 66 per arm. Splitting α across an either-wins pair (the 0.025/0.025 convention) halves the level of every test while licensing only the weak disjunctive claim that *something* differed. A composite of accuracy and time would have no substantive scale. The hierarchy preserves the theoretical pairing—both outcomes are estimated, reported, and interpreted together—while giving the confirmatory claim a single unambiguous target.

0.2.2 Primary accuracy model: estimand anchoring, model choice, covariates, and diagnostics

The mixed-model formulation of Equation (1) is chosen over the two natural alternatives. Against *separate per-wave analyses* (one ANCOVA-type model per wave): the single fit expresses the immediate-wave contrasts and their delayed-wave change as parameters of one model, so the persistence hypothesis (H_{5a}) is a within-model linear combination ($\beta_2 + \beta_5$) rather than a comparison across separately fitted models, and participants who provide only the immediate wave still contribute to all fixed effects through the likelihood (Bell & Rabe, 2020; Detry & Ma, 2016). With Wave dummy-coded (immediate = 0), the arm main effects are anchored at the immediate assessment and so coincide with the primary estimand, the pre-specified target of estimation (International Council for Harmonisation, 2019); the per-wave W. Lin (2013) robustness estimator (Appendix 0.2.7) retains a fully assumption-light single-wave check. Against *repeated-measures ANOVA*: RM-ANOVA discards any participant missing a wave (unbiased only under the far stronger missing-completely-at-random assumption), accommodates continuous prognostic covariates awkwardly, and yields omnibus wave-by-arm F -tests rather than the directional, wave-anchored contrasts the hypotheses specify. The mixed model for repeated measures is the standard confirmatory choice in trials with post-randomisation repeated outcomes precisely because it uses all available data and remains valid under missing-at-random given the modelled variables (Bell & Rabe, 2020; Detry & Ma, 2016).

With two waves, the participant random intercept is equivalent to a compound-symmetry covariance with a common residual variance. This is a restriction, not merely a

convenience, and two pre-specified guards would surface any distortion from wave-specific variances: the per-wave W. Lin (2013) estimator (which imposes no cross-wave structure at all) and the item-level crossed-random-effects model (Appendix 0.2.7, item 4). An unstructured covariance—the conventional MMRM default (Mallinckrodt et al., 2008), which with only two waves adds a single wave-specific variance parameter—is pre-specified as a sensitivity fit and reported in place of the compound-symmetric model if a likelihood-ratio test rejects the equal-variance restriction; because the primary immediate-wave contrasts are point-identified independently of the cross-wave covariance, the covariance structure bears only on the precision of the persistence estimand. Attrition concentrates at the delayed wave, so the primary immediate-wave contrasts are essentially complete-data by design; the missing-at-random assumption bears on the persistence estimand and is probed by the multiple-imputation and Lee-bounds sensitivity analyses (Missing data and attrition).

Adjustment for the two prognostic baseline covariates is preferred to an unadjusted comparison because it increases precision without biasing the estimate under randomisation. ANCOVA-type adjustment dominates unadjusted and change-score analyses in randomised studies (van Breukelen, 2006), re-analyses of completed trials show substantial realised power gains from prognostic adjustment (Kahan et al., 2014), and fitting baseline measurements as covariates is the standard recommendation for trials of this structure (Vickers & Altman, 2001) and is endorsed by regulatory guidance for confirmatory trials (European Medicines Agency, 2015; U.S. Food and Drug Administration, 2023). Both covariates were specified in the trial plan and are fitted regardless of their observed baseline distribution; following Senn (1994), no significance tests of baseline balance are conducted. All single-wave linear models in this plan (the robustness analyses and the productivity block, where each participant contributes one observation) use HC3 heteroskedasticity-consistent standard errors (Long & Ervin, 2000).

Assumption diagnostics. Residual and random-intercept normality are inspected with quantile–quantile plots. The bounded outcome (a proportion over 16 items) is guarded by the fractional-logit robustness model (Papke & Wooldridge, 1996) and by the item-level logistic specification, both of which respect the outcome’s support; a ceiling-compressed

accuracy distribution would surface there. The linear specification is nonetheless retained as the primary model on three grounds: it targets the marginal mean difference in percentage points—the directly interpretable, pre-registered effect scale; the ordinary-least-squares estimator remains consistent for that average treatment effect under randomisation despite the bounded, non-normal outcome (Angrist & Pischke, 2009); and at the assumed control mean of 0.60 over sixteen items the distribution sits well away from either bound, where the Gaussian approximation is close. A pre-registered simulation of this exact estimator on binomial-granularity outcomes returned nominal Type I error, and the fractional-logit and item-level logistic fits are reported alongside the linear model wherever they disagree. Wald inference from the REML fit relies on the large-sample normal approximation, defensible at $N = 200$; the pre-registered randomisation inference (Appendix 0.2.7, item 5) supplies a backstop whose validity derives from the physical act of randomisation alone (Young, 2019).

0.2.3 Key-secondary success-time model: estimand, competing risks, clustering, and diagnostics

Estimand. The event is the submission of a correct answer; incorrect, “I do not know”, and unanswered items are censored at their recorded response time. The arm coefficients are therefore *cause-specific* log hazard ratios for correct production: the instantaneous rate at which a still-unresolved item is converted into a correct answer. This is the survival-analysis expression of fluency, and it is the appropriate estimand here because the competing destination—an incorrect submission—is not a nuisance process to be marginalised over but the very phenomenon the primary endpoint measures. The competing-risks alternative, the Fine–Gray subdistribution hazard (Austin et al., 2016), targets the cumulative incidence of correct answers and therefore re-imports accuracy into the time endpoint, duplicating Equation (1) in a less interpretable metric; it is not used. The naive alternative—comparing mean response times among correctly answered items only—conditions on a post-treatment variable and is biased when correctness itself responds to treatment (Montgomery et al., 2018); the participant-level median time-to-correct and log rate correct score (Appendix 0.2.7, item 7) are retained only as trans-

parency checks that do not replace the censored formulation.

Inference and diagnostics. Because items are nested in participants, inference uses the robust sandwich covariance estimator clustered at the participant level, which remains valid under within-cluster dependence and model misspecification (D. Y. Lin & Wei, 1989). Ties are handled with the Efron approximation. The proportional-hazards assumption is assessed with scaled Schoenfeld residuals and the associated score test (Grambsch & Therneau, 1994); if it is violated for the arm indicators, the pre-specified fallback is a log-normal accelerated failure time model reporting time ratios, with the Cox estimates retained as descriptive. Response time is measured from stimulus onset to answer submission; no administrative cap is applied, following the pre-specified rationale that extreme values are handled by the speeding-exclusion sensitivity check (item 8 below) rather than by censoring. One caveat is stated rather than hidden: censoring an item at the moment of an incorrect submission is informative if fast-but-wrong responders differ systematically from slow responders. The cause-specific estimand remains well defined under this regime, and the scoring-sensitivity and speeding-exclusion analyses (Appendix 0.2.7, items 6 and 8) bound the influence of fast incorrect responding on the conclusions.

0.2.4 Multiplicity control, gatekeeping, and the non-inferiority test

Why a fixed sequence. The confirmatory hypotheses are ordered a priori by theoretical priority: $H1_a$ (outsourcing impairs), then $H2_a$ (scaffolding is not harmful), then the Socratic–Unrestricted supporting contrast. Testing each at level α and stopping at the first non-rejection controls the familywise error rate in the strong sense: a type I error can occur only at the first true null hypothesis in the sequence, and that hypothesis is only reached through rejections of the hypotheses ordered before it, each tested at level α (Bauer, 1991; Dmitrienko & D’Agostino, 2013; Maurer et al., 1995; U.S. Food and Drug Administration, 2017; Westfall & Krishen, 2001). The procedure spends no multiplicity tax on the primary contrast, whereas a flat Holm correction across the three contrasts (Holm, 1979) would control the same error rate while testing the primary contrast at $\alpha/3$ in the worst case—buying protection the a-priori ordering already provides, at the

price of power on the one contrast the trial is designed around. The structural cost of the sequence is accepted by design: if $H1_a$ is not rejected, $H2_a$ is never formally tested and the non-inferiority claim is forfeit even if true. This is theoretically coherent: the scaffolding question—does withholding the answer protect learning?—presupposes that unscaffolded answer provision harms it; absent that harm, $H2_a$ loses its comparator logic. The graphical α -recycling alternative (Bretz et al., 2009), which would allow $H2_a$ to be tested after an $H1_a$ failure by splitting α across branches, was considered and rejected because it taxes the primary contrast, and protecting the primary at full α is the design priority. The ordering is fixed before data collection and is never reordered on observed p -values, a requirement for the validity of the procedure.

The non-inferiority test. $H2_a$ asserts that Socratic scaffolding is not harmful relative to search within a tolerated margin: $H_0: \beta_1 \leq -\Delta$ against $H_1: \beta_1 > -\Delta$. This is the operative one-sided test of the two one-sided tests (TOST) procedure (Lakens, 2017; Lakens et al., 2018; Schuirmann, 1987); operationally, non-inferiority is declared if and only if the lower limit of the two-sided 90% confidence interval for β_1 lies above $-\Delta$, and the interval is reported whatever the verdict so that an underpowered result remains informative as an estimate (Amrhein et al., 2019; Cumming, 2014). The dedicated test exists because its naive substitute is invalid: a non-significant superiority test cannot establish absence of harm—absence of evidence is not evidence of absence (Altman & Bland, 1995)—whereas the margin test can positively reject harm larger than Δ .

The margin. A non-inferiority margin must be pre-specified and justified on substantive grounds, never widened to make power feasible (Althunian et al., 2017; European Medicines Agency, 2005; Piaggio et al., 2012). The margin here follows the preservation-of-effect logic (Althunian et al., 2017; D’Agostino et al., 2003): the reference effect is the deficit the trial is designed to detect, the ≈ 17 -percentage-point Control–Unrestricted difference anchoring $H1$ (Shen & Tamkin, 2026), and the margin requires the Socratic arm to preserve at least 50% of the control’s advantage over the harmful comparator, giving $\Delta = 0.5 \times 17 \approx 8.5$ percentage points. The margin thus sits well below the superiority anchor (coherence between the two claims), and the non-inferiority comparison is separately powered against it—approximately 0.87 at a true Socratic–Google difference

of zero—in Appendix ??.

Analysis population for the non-inferiority claim. Intention-to-treat is not conservative for non-inferiority: noncompliance and cross-arm contamination blur differences between arms and therefore push toward spurious equivalence. Following the CONSORT extension for non-inferiority trials (Piaggio et al., 2012), the H_{2a} conclusion requires consistency between the intent-to-treat and per-protocol estimates (Compliance, contamination, and complier effects), with any discrepancy reported and interpreted against the logged compliance data.

Secondary and lower families. The two secondary confirmatory accuracy hypotheses—the Bloom-order interaction (H_{4a}) and delayed-wave persistence (H_{5a})—form a Holm family at familywise $\alpha = 0.05$, one-sided (Holm, 1979). The key-secondary success-time contrasts form a separate Holm family interpreted as supporting evidence, never as substitutes for the primary conclusion. The aided-block productivity indicators are controlled at a false discovery rate (Benjamini & Hochberg, 1995) as converging indicators. Exploratory quantities (H_{4b} , H_{5b} , the four-level Bloom trend, moderation analyses) are reported as estimates with confidence intervals and carry no error-rate claim (frumento2025).

0.2.5 Secondary analysis I: Bloom-level concentration (full specification)

The four administered Bloom levels are treated as an *ordered* gradient of element interactivity, Remember < Understand < Analyze < Evaluate, whose ordering—but not spacing—is licensed by the taxonomy and corroborated empirically (Anderson & Krathwohl, 2001; Krathwohl, 2002; Srinivas et al., 2025).

The binary lower- versus higher-order partition of Equation (3) is preferred to a finer level-by-level analysis on three grounds. First, the lower- versus higher-order cut is the theoretically load-bearing distinction and is itself uncontested (Anderson & Krathwohl, 2001), whereas whether each adjacent level is strictly more demanding than the last is disputed (Hamamoto Filho et al., 2020; Ravand et al., 2025); the contrast rests only on the former. Second, with only four items per level per wave, an eight-versus-eight contrast is materially better powered than a four-point trend. Third, the closest published precedent—a four-arm comparison of GenAI, web search, an e-textbook, and a no-tool

control across Bloom-stratified tasks—recovered this pattern with a coarse per-level analysis: the tool advantage was pronounced for lower-order and application items and absent at the highest, analysis-and-evaluation level, a *step* at the lower- to higher-order boundary rather than a smooth gradient (Akgun & Toker, 2025). Throughout, whether the deficit *differs* across order levels is adjudicated by the interaction term itself, never by contrasting a significant effect at one level against a non-significant effect at another—an inference that is not valid (Gelman & Stern, 2006; Nieuwenhuis et al., 2011). Because interactions on the log-odds scale need not correspond to interactions on the probability scale (Ai & Norton, 2003), the focal arm $\times$ order contrast is additionally reported as an average marginal interaction contrast on the probability scale. The confirmatory interaction is estimated by fitting the mean structure of Equation (3) as a participant-clustered logistic GEE (exchangeable working correlation), whose one-sided test is well calibrated in simulation; the crossed-random-effects Bayesian GLMM is reported alongside as corroboration only, because its variational posterior standard deviations are anticonservative and would overstate the interaction’s precision.

As a pre-specified but supportive refinement, the full four-level ordering is examined with the arm $\times$ linear-polynomial interaction and a Jonckheere–Terpstra test against the monotone alternative, which assumes ordering without spacing (Jonckheere, 1954). Because interaction tests are markedly less efficient than tests of main effects (McClelland & Judd, 1993) and the per-level cells are thin, this trend is reported with point estimates and confidence intervals emphasised over significance verdicts (**frumento2025**) and treated as corroboration of, not a substitute for, the binary result.

Two analyses are exploratory. First, whether the higher-order concentration appears at acquisition, at retention, or both is examined by adding the wave factor and the arm $\times$ order $\times$ wave interaction to Equation (3), the item-level analogue of the persistence analysis; the precedent above found tool advantages that were positive immediately yet eroded by a delayed test (Akgun & Toker, 2025), so the deficit may surface at the delayed wave rather than the immediate one. As three-way interactions on four-item cells are severely underpowered, this is read from estimates and intervals alone. Second, should lower-order *sparing* be claimed, the Unrestricted–Search difference among the lower-order

items is assessed for equivalence against the pre-registered equivalence bound of 6.25 percentage points (one Mini-VLAT item) in each wave (Lakens, 2017; Lakens et al., 2018)—an equivalence threshold distinct from the non-inferiority preservation margin Δ (Appendix 0.2.4)—so that a claim of protection rests on a positive equivalence test rather than a non-significant difference. The success-time analogue of the moderation hypothesis (H_{4b}) is likewise exploratory: order-specific hazard ratios are reported with confidence intervals, without significance verdicts. The assumed level ordering is validated within each wave by confirming that mean item difficulty rises across the four levels, which doubles as a parallel-form check.

0.2.6 Secondary analysis II: productivity (frontier plot and affordance check)

The rate correct score is analysed on the log scale because its sampling distribution is strongly right-skewed, and the aided productivity outcomes of Equation (4) are controlled at a false discovery rate (Benjamini & Hochberg, 1995) rather than individually because they function as converging indicators rather than confirmatory claims. The productivity–learning trade-off itself is summarised descriptively as an arm-level frontier plot of aided throughput against the immediate-wave unassisted accuracy of Equation (1); the confirmatory weight of the trade-off claim rests on the primary accuracy endpoint, not on this plot. These analyses simultaneously serve as the affordance manipulation check: a markedly higher aided throughput and aided success-hazard in the Unrestricted arm confirms that the arms differed in solution-provision capability.

0.2.7 Robustness analyses

The primary and key-secondary conclusions should be robust to the following pre-specified perturbations.

1. **Unadjusted estimates.** Difference-in-means (accuracy) and an unadjusted Cox model (success time), demonstrating independence from the covariate model.
2. **Fully interacted adjustment.** The W. Lin (2013) estimator applied to each wave’s accuracy separately: treatment indicators, mean-centred covariates, and all treatment–covariate interactions with sandwich standard errors.

3. **Bounded-outcome specification.** A fractional logit model per wave estimated by quasi-maximum likelihood (Papke & Wooldridge, 1996), reporting average marginal effects, guarding against ceiling-effect distortion in the linear model.
4. **Item-level accuracy model.** A logistic mixed-effects model on item-level correctness with crossed random intercepts for participants and items (Baayen et al., 2008), with arm and wave as fixed effects, verifying that participant-level aggregation masks no item heterogeneity.
5. **Randomisation inference.** Permutation p -values for the primary contrast in both the accuracy and success-time models, based on 10,000 re-randomisations of arm labels with the full estimation pipeline re-run per draw, following evidence that conventional asymptotic tests can overstate significance in experimental samples of this size (Young, 2019).
6. **Scoring sensitivity.** Accuracy re-estimated with “I do not know” responses (a) scored as incorrect, the primary coding, and (b) excluded so accuracy is computed over attempted items. The success-time endpoint is structurally invariant to this choice, since “I do not know” responses are censored under either convention.
7. **Simple success-time summaries.** Participant-level median time-to-correct and the log rate correct score on the unassisted blocks, analysed in the linear framework, as transparency checks on the Cox result.
8. **Data-quality exclusions.** Re-estimation excluding participants flagged by the pre-registered speeding rule (median item response time below 1,000 ms) or failing the embedded attention check, applied symmetrically across arms. This rule also limits the influence of inattentive responding on all response-time outcomes.

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